Auditoría, Calidad y Fiabilidad Informáticas

(IT Auditing, Quality, and Reliability)

Universidad Complutense de Madrid Academic Year 2026/27

Practical 3: Peg solitaire

Peg solitaire (*Senku* in Spanish) is a single-player game that can be played on different kinds of boards. At the start, every position but one holds a peg, and the goal is to end up with only one. To get there, pegs jump over one another, "capturing" the peg they jump over, as in draughts. More information is available here. In this practical we will work out how to solve the triangular version of peg solitaire in Maude.

Exercise 1 Design a data type to represent a peg solitaire board. We are interested in particular in the triangular board, although you may make the data structure as general as you like.

Exercise 2 Implement the jumps of the pegs as rules.

Exercise 3 Define an initial triangular board with 5 rows that has a single hole in the second position of the third row (counting from the top), and use the `search` command to find:

(a) Any solution.

(b) The "perfect" solution, the one in which the only remaining peg is in the middle of the third row of the triangle, counting from the top (that is, where we initially placed the hole).